Osmary A. Medina-Báez
Department of Ecology and Evolutionary Biology
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EDUCATION

2020 – 2025 | Ph.D. in Biology, Case Western Reserve University, Cleveland, OH
Dissertation: Vulnerability of butterflies to climate change
Advisor: Dr. Karen C. Abbott
Co-advisor: Dr. Jean H. Burns
2018 – 2020 | M.Sc. in Biology, John Carroll University, University Heights, OH
Thesis: Are populations of the salamander *Bolitoglossa altamazonica* declining at low elevations due to rising temperatures?
Advisor: Dr. James I. Watling
2011 – 2016 | B.Sc. in Biology, University of Puerto Rico, Mayagüez, PR

PUBLICATIONS

* Undergraduate student

Medina-Báez OA, A Lenard, RA Muzychuk*, CRB da Silva, and SE Diamond. 2023. Life cycle complexity and body mass drive nonlinear climate vulnerabilities across ontogeny. *Conservation Physiology* 11 (1), coad058

Veselka AJ, A Aponte-Gutiérrez, **OA Medina-Báez** and JI Watling. 2023. Upper Thermal Limits Predict Herpetofaunal Responses to Forest Edge and Cover in Tropical Forest. *Biotropica*, *55*(2), pp.540-550.

Prileson EG, J Clark*, SE Diamond, A Lenard, **OA Medina-Báez**, and RA Martin. 2023. Keeping your cool: No evidence for an evolutionary loss in overwintering physiology in response to urbanization in the Acorn Ant, *Temnothorax curvispinosus*. *Journal of Thermal Biology* 114, 103591

Fair H, **OA Medina-Báez**, BJ Spiecker, Q Gan, YYJ Cheung, E D'Bastiani, GR Goldsmith. 2024. Can AI interpretation increase inclusivity? *Frontiers in Ecology and the Environment*, *22*(10), e2821.

Diamond SE**,** CRB da Silva and **OA Medina-Báez**. 2024. A multicontinental dataset of butterfly thermal physiological traits. *Scientific data*, *11*(1), 1348.

Santiago-Rosario LY, AM Alonso-Rodríguez, **OA Medina-Báez**, G Errico, T Lubin, T, CdM Valle-Rodríguez, & L Jiménez. 2026. From Research to Action: Communicating Science Effectively for Real-World Impact. *Bulletin of the Ecological Society of America*, *107*(2), 1-6.

GRANTS AND AWARDS

2026 | **SEEDS Graduate Alumni Travel Award** | all expenses paid | ESA Annual Meeting 2026
2025 | **SEEDS Graduate Alumni Travel Award** | all expenses paid | ESA Annual Meeting 2025
2025 | **Honorable Mention, ESA Latin America and the Caribbean Chapter Poster Award |** ESA Annual Meeting 2025
2025| **Best student talk at Biology Graduate Symposium** | $100 | Case Western Reserve University, Department of Biology
2024 | **International Conservation Fund of Canada Research Fellowship |** $10,000 **|** Co-PI **|** Manu Biological Station, Perú
2024 | **Ruth Barber Moon Graduate Student Award |** $1,000 **|** Case Western Reserve University
2024 **| Oglebay Foundation Research Small Grants |** $1,300 **|** PI **|** Case Western Reserve University Department of Biology Oglebay Fund
2024 | **School of Graduate Studies Graduate Student Travel Award** | $500 | Case Western Reserve University
2024 | **Oglebay Foundation Travel Small Grants |** $1,500 **|** Case Western Reserve University Department of Biology Oglebay Fund
2021 | **Oglebay Foundation Travel Small Grants |** $1,071 **|** Case Western Reserve University Department of Biology Oglebay Fund
2018 | **Segal Education Award** | $1,222.22 **|** AmeriCorps

INVITED PRESENTATIONS

Medina-Báez OA. Lo que revelan los insectos sobre un planeta cambiante. Con Ciencia en las Américas Talk series, Latin America and the Caribbean Chapter of the Ecological Society of America. YouTube: https://youtube.com/live/PVSP7xX4rJo?feature=share

Medina-Báez OA. Vulnerability of butterflies to climate change**,** Biology seminar, March 18, 2025, John Carroll University, University Heights, OH. (oral presentation)

Medina-Báez OA. Vulnerabilidad de las mariposas ante el cambio climático [ENG Vulnerability of butterflies to climate change]. Biology Association talk series, October 24, 2024, University of Puerto Rico, Mayaguez, PR. (oral presentation, remote, Spanish)

Medina-Báez OA. Assessing vulnerability of butterflies to climate change in Ohio using their physiological tolerances. Board of trustees meeting, October 24, 2024, Case Western Reserve University, Cleveland, OH. (oral presentation)

Medina-Báez OA, RA Muzychuk*, GG Deme, KC Abbott, JH Burns. *¿Qué herramientas ayudan ante el cambio?* Physiological tolerances as a tool to assess butterfly vulnerability to climate change. Organized Special Session: Towards Equity in the Communication of Science: Harnessing the Power of AI for an Inclusive Tomorrow. August 7, 2024. Ecological Society of America Annual Meeting, Long Beach, CA. (oral presentation, Spanish with English AI interpretation)

Medina-Báez OA. Can physiological traits predict butterfly vulnerability to climate change? April 30, 2024. Cuyahoga Valley National Park Pollinator Symposium, Peninsula, OH (oral presentation)

Medina-Báez OA. Using R Studio to create your personal website. Website design workshop & social. Women in Science and Humanities Earning Doctorates (WISHED) association, April 11, 2024. Case Western Reserve University, Cleveland, OH (oral presentation)

Medina-Báez OA. Efecto del cambio climático sobre organismos ectotermos [ENG *The effects of climate change in ectotherms*]. March 10, 2022. Women in Science Seminar Series. University of Puerto Rico, Arecibo, PR (oral presentation, remote, Spanish)

Medina-Báez OA. El efecto del cambio climático sobre la salamandra *Bolitoglossa altamazonica* en Villavicencio [ENG *The effects of climate change in the salamander Bolitoglossa altamazonica in Villavicencio*]. May 12, 2021. Talk series "Hablemos de Anfibios y su conservación", Secretaría del Medio Ambiente, Villavicencio, Colombia (oral presentation, remote, Spanish)

ORGANIZED SESSIONS AND SYMPOSIA

2026 | **Medina-Báez OA, CdM Valle-Rodríguez, SJ Tumber-Dávila.** Ecosystem Responses to Drought in the Neotropics. July 28. Ecological Society of America Annual Meeting, Salt Lake City, UT. (Organizer & moderator; Organized Oral Session)

2024 | H Fair, **OA Medina-Báez**, B Spiecker, HG Ortiz Cano. Towards Equity in the Communication of Science: Harnessing the Power of AI for an Inclusive Tomorrow. August 7. Ecological Society of America Annual Meeting, Long Beach, CA. (Organizer & speaker; Special Session)

2023 | **Medina-Báez OA**, HG Ortiz Cano, B Girdler. ¿Cómo haces Ecología? Como você faz Ecologia? How do you do Ecology? Latine Voices in Our Community of Ecologists. August 8. Ecological Society of America Annual Meeting, Portland, OR. (Organizer and moderator; Inspire Session)

PRESENTATIONS

* Undergraduate student

† High school student

Medina-Báez OA, AT Killeffer, OT White, KS Sheldon. Experimental increases in temperature mean and variance alter reproductive behaviors and offspring fitness in two native dung beetles. April 11, 2026. Ecological Society of America Great Lakes Regional Conference, Cleveland, OH (oral)

Sexton E*, **OA Medina-Báez**, OT White, AT Killeffer & KS Sheldon. Behavioral responses to temperature change in the dung beetle *Phanaeus vindex.* September 27, 2025. The Southeastern Conference for Animal Behavior, University of Tennessee, Knoxville, TN

Medina-Báez OA, JH Burns, KC Abbott. Projected habitat suitability for specialist North American butterflies and their larval hostplants. August 11, 2025. Ecological Society of America Annual Meeting, Baltimore, MD. (poster)

Medina-Báez OA. Assessing vulnerability of butterflies to climate change in Ohio using their physiological tolerances. February 28, 2025. Biology Department Graduate Symposium, Cleveland Botanical Garden, Cleveland, OH. (oral)

Chen W*, A Pandian†, **OA Medina-Báez**, JH Burns. Analysis of Butterfly Wing Area and Coloration Using the Specialized Computer Program WinFolia. August 9, 2024. Symposium for Ecological Research, Cleveland Botanical Garden, OH. (poster)

Medina-Báez OA, RA Muzychuk*, GG Deme, SE Diamond. Assessing vulnerability of butterflies to climate change in Ohio using their physiological tolerances. April 6, 2024. Ecological Society of America Great Lakes Regional Conference, Kalamazoo, MI (oral)

Medina-Báez OA, RA Muzychuk*, SE Diamond. Can butterfly thermal physiological traits predict their vulnerability to climate change? December 12, 2023. Cleveland Metroparks Natural Resources Research Symposium, Parma, OH. (poster)

Medina-Báez OA, RA Muzychuk*, SE Diamond. Can butterfly thermal physiological traits predict their vulnerability to climate change? August 10, 2023. Ecological Society of America Annual Meeting, Portland, OR. (poster)

Diamond SE, **OA Medina-Báez,** CRB da Silva. May 16, 2023. Thermal physiological traits mediate butterfly range shifts under recent climate change. Species on the Move Conference, Bonita Springs, FL. (oral)

Medina-Báez OA, A Lenard, RA Muzychuk*, CRB da Silva, and SE Diamond. Life cycle complexity drives nonlinear, size-decoupled climate vulnerabilities across ontogeny. August 18, 2022. Ecological Society of America Annual Meeting, Montréal, Canada. (poster)

Diamond SE, CRB da Silva and **OA Medina-Báez.** Can species trait-based models of contemporary range shifts be rescued? June 28, 2022. Society for the Study of Evolution Annual Meeting, Cleveland, Ohio. (oral)

Medina-Báez OA and SE Diamond. Do thermal physiological traits explain variation in contemporary, climate-driven range shifts in North American butterflies? January 3-7, 2022. Society for Integrative and Comparative Biology Annual Meeting, Phoenix, AZ. (oral, remote)

Muzychuk RA*, **OA Medina-Báez**, and SE Diamond. Interspecific variation in butterfly warming tolerances and consequences for population size under recent climate change. January 3-7, 2022. Society for Integrative and Comparative Biology Annual Meeting, Phoenix, AZ. (poster)

Muzychuk RA*, **OA Medina-Báez**, and SE Diamond. Interspecific variation in butterfly warming tolerances and consequences for population size under recent climate change. Dec 3 – Dec 10, 2021. Case Western Reserve University. Intersections: SOUCRE symposium. (poster)

Pu X*, A Lenard, **OA Medina-Báez**, and SE Diamond. Thermal limits across ontogeny in the butterfly *Vanessa cardui*. Dec 3 – Dec 10, 2021. Case Western Reserve University. Intersections: SOUCRE symposium. (poster)

Medina-Báez OA, AF Aponte-Gutiérrez, AJ Veselka, and JI Watling. Are populations of the salamander *Bolitoglossa altamazonica* declining at low elevations due to rising temperatures? January 5, 2021. Society for Integrative and Comparative Biology Annual Meeting, Virtual Conference. (oral, remote)

Medina-Báez OA and JI Watling. October 26, 2019. Thermal safety margins of *Bolitoglossa altamazonica* across an altitudinal gradient. OU/JCU Symposium, Ohio University, Athens, Ohio. (oral)

Medina-Báez OA and JI Watling. October 20, 2018. Critical Thermal limits in *Bolitoglossa altamazonica* across an elevational gradient. OU/JCU Symposium, John Carroll University, University Heights, Ohio. (oral)

Medina-Báez OA, KJ Hovey and KM O'Donnell. Breeding season phenology and demography of Frosted Flatwoods Salamanders (*Ambystoma cingulatum*) at St. Marks National Wildlife Refuge. March 24-25, 2018. 41st Annual Herpetology Conference, Gainesville, Florida. (poster)

Hovey KJ**, OA Medina-Báez**, KM O'Donnell, W Barichivich and S Walls. Breeding season phenology and demography of Frosted Flatwoods Salamanders (*Ambystoma cingualatum*) at St. Marks National Wildlife Refuge. February 22-25, 2018. Southeast Partners in Amphibian and Reptile Conservation Meeting, Helen, Georgia. (poster)

Medina-Báez OA and AJ Novarro. Which species is better equipped for climate change? Thermal limits in two widely distributed salamanders. January 5, 2017. Society for Integrative and Comparative Biology Annual Meeting, New Orleans, Louisiana. (poster)

Medina-Báez OA* and AJ Novarro. Which species is better equipped for climate change? Thermal limits in two widely distributed salamanders. July 29, 2016. Mountain Lake Biological Station REU Seminar, Pembroke, Virginia. (oral)

Medina-Báez OA* and M Vega. General information about the invasive snake *Boa constrictor* in Puerto Rico. April 22, 2015. ARMY Earth Day Expo**.** Fort Buchanan, Guaynabo, Puerto Rico. (poster)

Medina-Báez OA* and MJ González. Effects of light and fish predation on benthic invertebrates. February 13, 2014. 7th NEA Science Day, University of Puerto Rico, Mayagüez, Puerto Rico. (poster)

Medina-Báez OA* and MJ González. Effects of light and fish predation on benthic invertebrates. August 2, 2013. Miami University REU Symposium, Oxford, Ohio. (oral)

CONTRIBUTED WORKSHOPS

2026 | **Organizer**, Workshop: From Research to Action: Communicating Science Effectively for Real-World Impact. **Medina-Báez OA**, Santiago-Rosario LY, Tumber-Dávila SJ, Prado-Valladares AC & Kandlikar G. Ecological Society of America Annual Meeting, Salt Lake City, UT. July 28

2025 | **Invited panelist**, Workshop: From Research to Action: Communicating Science Effectively for Real-World Impact. Topic: Sharing your research as an early-career academic. Santiago-Rosario LY, A Alonso-Rodríguez, **OA Medina-Báez**, G Errico, T Lubin, CDM Valle-Rodríguez, and L Jiménez. Ecological Society of America Annual Meeting, Baltimore, MD, USA. August 12

2023 | **Co-organizer**, Introduction to Innovations Workshop, Butterfly metabolic rate. Diamond SE, **OA Medina-Báez** and GG Deme. Case Western Reserve University, Cleveland, OH. July 21

2022 | **Co-organizer**, Introduction to Innovations Workshop, Butterfly life cycle. Diamond SE, **OA Medina-Báez** and E Prileson. Case Western Reserve University, Cleveland, OH. July 22

RESEARCH AND WORK EXPERIENCE

2025 - present | **The University of Tennessee, Knoxville,** Department of Ecology & Evolutionary Biology, Knoxville, TN
NSF Postdoctoral Research Fellow, Sheldon Lab
UTK Postdoctoral Research Associate, Sheldon Lab

2020 – 2025 | **Case Western Reserve University,** Department of Biology, Cleveland, OH
Graduate Research, Abbott Lab, Burns Lab & Diamond Lab
Research Assistant, Diamond Lab
Research topic: Butterfly thermal physiology, habitat suitability modeling and conservation

2018 – 2020 | **John Carroll University,** Biology Department, University Heights, OH
Graduate Research, Watling Lab
Research topic: Tropical salamander thermal physiology and population decline

2017 – 2018 | **United States Geological Survey** (USGS), St. Marks Natl. Wildlife Refuge, St. Marks, FL
Environmental Steward, Conservation Legacy (AmeriCorps) and USGS, October – May
Research topic: Breeding, phenology and conservation of the threatened frosted flatwood's salamander *Ambystoma cingulatum*

2017 | **Biomarine Professional Services**
Manatee Observer, San Juan Harbor Maintenance Dredge, San Juan, Puerto Rico, April – July
Job: Ensure that dredging operations did not harm or endanger manatees

2016 | **Mountain Lake Biological Station,** University of Virginia, Pembroke, VA
Undergraduate Researcher, NSF Research experience for Undergraduates (REU) Program, May – July
Research topic: Thermal limits across elevation in two widely distributed salamanders

2015 – 2017 | **University of Puerto Rico,** Biology Department, Mayagüez, PR
Undergraduate Field Assistant, Data and specimen collection
Research topic: Niche partitioning strategies of two endemic *Sphaerodactylus* species found at Salt Flats Refuge in Cabo Rojo, P. R.
Undergraduate Field and Research Assistant, **Telemetry data** and specimen collection and dissections of *Boa constrictor*. Outreach for general audiences, offering talks about Herpetofauna in Puerto Rico

Research topic: Movement and habitat distribution of an invasive species (*Boa constrictor*) in Puerto Rico

2015 – 2016 | **University of Puerto Rico**, Aguadilla, PR

Technician for the West Coast Sea Turtle Project

Job: Monitoring sea turtle's nests, hatchings, species management. Outreach for schools and the general public offering talks about sea turtle conservation.

2015 | **Field stations in Guánica, Cabo Rojo and Vieques,** PR

Volunteer Field and Research Assistant, Data and specimen collection for PhD dissertation of N Angeli at Texas A&M. Practiced specimen preservation and tissue sample collection of *Pholidoscelis wetmorei* and *P. exsul.* Summer

2013 | **Miami University**, Oxford, OH

Undergraduate Researcher, NSF Research experience for Undergraduates (REU) Program, Summer (10 weeks)

Research topic: Effect of light and fish predation on benthic invertebrates

TEACHING EXPERIENCE

2026 | **University of Puerto Rico, Ponce,** Biology Department, Ponce, PR

Co-instructor, Tropical Terrestrial Ecosystems Ecology laboratory course, Spring semester

Topics: Introduction to GIS, QGIS software, spatial data analysis

2020 – 2024 | **Case Western Reserve University,** Biology Department, Cleveland, OH

2025 | *Guest Lecturer,* Foundations of Advanced Ecology course, February

Topic: Global change

2024 | *Guest Lecturer*, Ecology course, September

Topic: Adaptations to variable environments

2024 | *Guest Lecturer*, Taming the tree of Life: Phylogenetic Comparative Methods, February

Topic: Introduction to Phylogenetic Generalized Least Squares Analysis

2024 | *Teaching Assistant*, Taming the tree of Life: Phylogenetic Comparative Methods, Spring

2020-2024 | *Teaching Assistant*, Genes, Ecology and Evolution Laboratory, 4 semesters

2024 | **LeMoyne-Owen College,** Division of Natural and Mathematical Sciences, Memphis, TN

Guest Lecturer, Methods in Biological Sciences, April 10

Topic: Ecophysiology of butterflies: which groups are most vulnerable to climate change?

2021 | **University of Puerto Rico, Arecibo,** Biology Department, Arecibo, PR

Guest Lecturer, General Biology course, Spring

Topic: Applying to NSF REU programs, and thesis seminar: Are populations of the salamander *Bolitoglossa altamazonica* declining at low elevations due to rising temperatures?

2018 – 2020 | **John Carroll University,** Biology Department, University Heights, OH.

2020 | *Graduate Assistant,* Geographic Information Systems Laboratory, Spring

2019 – 2020 | *Graduate Assistant,* Principles of Biology Laboratory II, Spring

2019 | *Graduate Assistant,* Principles of Biology Laboratory III, Fall

2019 | *Graduate Assistant,* Environmental Earth Science Laboratory, Fall

2018 | *Graduate Assistant,* Principles of Biology Laboratory I, Fall

MENTORSHIP

2026 – present | **Research mentor**, University of Puerto Rico, Ponce, PR

Manuel Ramos-Feliciano, undergraduate student, Spring 2026-present

Leonardo Rodríguez-Dávila, undergraduate student, Spring 2026-present

2026 | **SEEDS Alumni Mentor**, Ecological Society of America Annual Meeting, Salt Lake City, UT

2025 – present | **Research mentor**, University of Tennessee, Knoxville, TN

Nathan Rhodes, undergraduate student, Spring 2026-present

Myles Nellans, undergraduate student, Spring 2026-present

Gracie Bixler, undergraduate student, Spring-Summer 2026

Emily Wagner, undergraduate student, Spring 2026

Emily Sexton, undergraduate student, Fall 2025-Spring 2026

2025 | **SEEDS Alumni Mentor**, Ecological Society of America Annual Meeting, Baltimore, MD
2020 – 2025 | **Research mentor**, Case Western Reserve University, Cleveland, OH
Wei Chen, undergraduate student, Summer 2024 – Spring 2025
Ahalya Pandian, high school student, Solon High School, Summer 2024
Rut A. Muzychuk, undergraduate student, Spring 2021-Fall 2023
Jessica Furlough, undergraduate student, Summer 2022
Melina Gabele, undergraduate student, Fall 2021 and Summer 2022
Xuan Pu, undergraduate student, Fall 2021
2023 – 2024 | **Mentor,** YES/SEO Near Peer Mentoring Program**,** Working with High School students navigating research and academia. Case Western Reserve University, Cleveland, OH. June 7 – July 28, 2023 & June 5 – July 24, 2024

SERVICE AND OUTREACH
2026 | **Application reviewer**, Ecological Society of America SEEDS student field trip, June 2026
2026 | **Abstract Reviewer,** Ecological Society of America Great Lakes Chapter Regional Conference, Cleveland, OH. Reviews: March 15, conference dates: Apr 10-12
2025 | **Application reviewer**, Ecological Society of America SEEDS student field trip, June 2025
2025 | **Invited discussion panelist**, LMSA and SACNAS Research Opportunities & Networking Event, Case Western Reserve University, Cleveland, OH. March 28
2024 | **Abstract Reviewer,** Ecological Society of America Great Lakes Chapter Regional Conference, Kalamazoo, MI. Reviews: March 14, conference dates: Apr 5-7
2023 | **Thesis Defense Committee Member** for undergraduate student Yeni T. Molina Moreno. Thesis topic: "Calidad y sensibilidad térmica de *Allobates juanii* (Morales, 1994) en diferentes coberturas de la vereda La Cuncia, (Villavicencio, Meta)". Universidad de los Llanos, Villavicencio, Meta, Colombia. October 26
2022 | **Thesis Proposal Committee Member** for undergraduate student Yeni T. Molina Moreno. Thesis topic: "Calidad y sensibilidad térmica de *Allobates juanii* (Morales, 1994) en diferentes coberturas de la vereda La Cuncia, (Villavicencio, Meta)". Universidad de los Llanos, Villavicencio, Meta, Colombia. March 17
2021 | **Moderator and Judge** for Virtual Intersections SORCE Symposium, Case Western Reserve University, Cleveland, OH. December 3
2020 | **Thesis Defense Committee Member** for undergraduate student Sandra C. Valderrama. Thesis topic: "Ecología del comportamiento térmico de *Hyloxalus aff. subpunctatus* (anura: dendrobatidae), en el páramo de Ocetá, Boyacá-Colombia" Universidad Pedagógica y Tecnológica de Colombia, Tunja, Boyacá, Colombia. December 14

LEADERSHIP ROLES
2025 – present |**Early career representative,** Ecological Society of America Latin America and the Caribbean Chapter
2025 – 2026 | **Multiple roles,** Planning committee, Ecological Society of America Great Lakes Chapter Regional Conference 2026, Cleveland, OH
2025- 2026 | **Social media manager,** Ecological Society of America Great Lakes Chapter
2024 - 2025 | **Student Representative**, Ecological Society of America Great Lakes Chapter
2024 - 2025 | **Executive board member**, Women in Science and Humanities Earning Doctorates (WISHED)
2023- 2024 | **Program Director,** Planning committee, Ecological Society of America Great Lakes Chapter Regional Conference 2024, Kalamazoo, MI

PROFESSIONAL DEVELOPMENT
2024 | **Entering mentoring** – An interactive short course on evidence-based practices for mentorship, August 4, Ecological Society of America Annual Meeting, Long Beach, CA.
2024 | **Generalized Linear Mixed Modeling Deep Dive:** Selecting conditional distributions, checking assumptions, and drawing inference for ecological data, August 9. Ecological Society of America Annual Meeting, Long Beach, CA.

2024 | **Ecological Society of America SEEDS GIS Workshop,** May 20-25, Florida A&M University, Tallahassee, FL
2023 | **Powerful Presentations - Speaking with impact,** Weatherhead Executive Education, May 23, Case Western Reserve University, Cleveland, OH
2020 | **Advanced Professional Development for University Teaching**, Fall semester, Case Western Reserve University, Cleveland, OH

PROFESSIONAL SOCIETIES

Ecological Society of America | SEEDS Alumni | Latin America and Caribbean Chapter | Great Lakes Chapter
Society for Integrative and Comparative Biology

LANGUAGES

English proficient level
Spanish proficient level (Native)

REFERENCES CONTACT INFORMATION

https://osmarymedina.github.io/

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Dr. Jean H Burns
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Dept. of Biology
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Dr. James I Watling
Master's advisor
Dept. of Biology
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